

When lookout sees crackle:

Anomaly detection via kernel density estimation

Rob J Hyndman¹, Sevvandi Kandanaarachchi^{2*}, Katharine Turner³

¹Monash University, Australia.

²CSIRO, Australia.

³Australian National University, Australia.

*Corresponding author(s). E-mail(s): sevvandi.kandanaarachchi@data61.csiro.au;

Abstract

We present an updated version of *lookout* – an algorithm for detecting anomalies using kernel density estimates with bandwidth based on Rips death diameters – with theoretical guarantees. The kernel density estimator for updated *lookout* is shown to be consistent, and the proposed multivariate scaling is robust and efficient. We show our updated algorithm performs better than the previous version on diverse examples.

Keywords: Bandwidth selection, Extreme value theory, Generalized Pareto distribution, Outlier detection, Persistent homology, Topological data analysis

1 Introduction

[Kandanaarachchi and Hyndman \(2022\)](#) proposed the *lookout* algorithm for anomaly detection using a kernel density estimate with bandwidth based on persistence homology. Anomalies are identified by fitting a Generalized Pareto Distribution (GPD) to the negative logarithm of the smallest leave-one-out kernel density estimates, and declaring points with low GPD probabilities as anomalies.

In this paper, we explore some of the theoretical properties of the *lookout* algorithm, particularly the choice of bandwidth based on the Rips death diameters, and the consistency of the kernel density estimates produced by this choice. This leads us to propose a modified *lookout* algorithm using a different bandwidth, although still based on the Rips death diameters. The modified algorithm improves the performance and robustness of the original algorithm, under a broader range of conditions.

First we introduce some background material, before describing both the original *lookout* algorithm, and our proposed modifications.

1.1 Anomalies

There is no broadly accepted definition for an anomaly. [Hawkins \(1980\)](#) defines an anomaly or outlier as *an observation which deviates so much from other observations as to arouse suspicion it was generated by a different mechanism*. [Barnett and Lewis \(1994\)](#) define an anomaly or outlier as *an observation (or a subset of observations) which appears to be inconsistent with the remainder of that set of data*. Both these definitions agree that anomalies are rare and different from other observations. Usually that is applied by fitting some (conditional) probability distribution to the set of observations, and considering whether each observation can be considered a likely random draw from the probability distribution. We formalize this in the following definition.

Definition 1 (Anomalies). Given a set of observations $\mathbf{y}_i \in \mathbb{R}^m$ for $i \in \{1, \dots, n\}$, and a probability density function f , we define observation $\mathbf{y}_i$ as anomalous if

$$f(\mathbf{y}_i) < \alpha, \tag{1}$$

for some small value of $\alpha > 0$.

1.2 Extreme Value Theory

Consider n independent and identically distributed random variables $X_1, \dots, X_n$ having a distribution function F , with their maximum $M_n = \max\{X_1, \dots, X_n\}$. Extreme Value Theory (EVT) analyses and characterizes the behavior of M_n . One of the fundamental results of EVT is the Fisher-Tippett-Gnedenko Theorem. We restate a version of it from [Coles \(2001, Theorem 3.1.1\)](#).

Theorem 1 (Fisher-Tippett-Gnedenko) *If there exist sequences of constants $\{a_n > 0\}$ and $\{b_n\}$ such that*

$$P\{(M_n - b_n)/a_n \leq z\} \rightarrow G(z) \quad \text{as } n \rightarrow \infty,$$

for a non-degenerate distribution function G , then G is a member of the Generalized Extreme Value (GEV) family

$$G(z) = \exp \left\{ - \left[1 + \xi \left(\frac{z - \mu}{\sigma} \right) \right]^{-1/\xi} \right\}, \tag{2}$$

where the domain of the function is $\{z : 1 + \xi(z - \mu)/\sigma > 0\}$ and the parameters $\mu, \xi \in \mathbb{R}$ and $\sigma > 0$.

The parameters μ , σ and ξ are called the location, scale and shape parameters respectively. The shape parameter ξ determines the behavior of the tails and differentiates between the Gumbel, Fréchet and Weibull distributions: a Fréchet distribution is obtained when $\xi > 0$, a Weibull distribution is obtained when $\xi < 0$, and a Gumbel distribution is obtained in the limit as $\xi \rightarrow 0$. Importantly, when F has a finite upper bound, then $\xi < 0$.

A second result from EVT that we will use is Pickands theorem (Pickands 1975), here restated from (Coles 2001, Theorem 4.1).

Theorem 2 (Pickands) *Let $X_1, \dots, X_n$ be a sequence of independent random variables with a common distribution function F , and let $M_n = \max\{X_1, \dots, X_n\}$. Suppose F satisfies Theorem 1, so that for large n , $P\{M_n \leq z\} \approx G(z)$, where*

$$G(z) = \exp \left\{ - \left[1 + \xi \left(\frac{z - \mu}{\sigma} \right) \right]^{-1/\xi} \right\},$$

for some $\mu, \xi \in \mathbb{R}$ and $\sigma > 0$. Then for large enough u ,

$$H(x) = \Pr(X \leq u + x \mid X > u) = 1 - \left(1 + \frac{\xi x}{\sigma_u} \right)^{-1/\xi}, \quad (3)$$

where the domain of H is $\{x : x > 0 \text{ and } (1 + \xi x)/\sigma_u > 0\}$, where $\sigma_u = \sigma + \xi(u - \mu)$.

The family of distributions defined by equation (3) is called the **Generalized Pareto Distribution** (GPD). The shape parameter ξ is the same in both GPD and GEV distributions. Furthermore, the GEV parameters can be recovered from the GPD formulation by estimating a GPD on all observations above some threshold.

1.3 Kernel Density Estimation

Given a set of observations $\mathbf{y}_i \in \mathbb{R}^m$ for $i \in \{1, \dots, n\}$, a kernel density estimate is defined as (Chacón and Duong 2018)

$$\hat{f}(\mathbf{y}) = \frac{1}{n} \sum_{i=1}^n |\mathbf{H}|^{-1/2} K(\mathbf{H}^{-1/2}(\mathbf{y} - \mathbf{y}_i)), \quad (4)$$

where K is a square-integrable spherically-symmetric function, bounded below by 0, with a finite second-order moment and unit integral, and $\mathbf{H}$ is a symmetric $m \times m$ positive-definite matrix.

Definition 2 (Consistency). Let $Y_1, \dots, Y_n$ be a sequence of independent random variables from a continuous distribution with square integrable density function f . Then (4) is a *consistent* estimator of f if

$$\lim_{n \rightarrow \infty} \mathbb{E} \left[\left(\hat{f}(\mathbf{u}) - f(\mathbf{u}) \right)^2 \right] = 0$$

The following result is provided by Chacón and Duong (2018, p31–37).

Theorem 3 *For $\hat{f}$ to be a consistent estimator, $\mathbf{H}$ must be a symmetric positive-definite matrix that satisfies the following conditions:*

- All elements of $\mathbf{H}$ approach zero as $n \rightarrow \infty$; and
- $n^{-1}|\mathbf{H}|^{-1/2} \rightarrow 0$ as $n \rightarrow \infty$.

We consider matrices of the form $\mathbf{H} = h_n \mathbf{I}_m$, where $h_n \in \mathbb{R}$ is a function of the sample $\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$

Definition 3. We say that h_n is an *admissible bandwidth* if $(h_n)^{2/m} \rightarrow 0$ and $nh_n \rightarrow \infty$ as $n \rightarrow \infty$ almost surely.

Thus, an admissible bandwidth leads to a consistent kernel density estimator.

1.4 The lookout algorithm for anomaly detection

The *lookout* algorithm of [Kandanaarachchi and Hyndman \(2022\)](#) has four inputs: the dataset $\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$, where each $\mathbf{y}_i \in \mathbb{R}^m$; a parameter $\beta \in (0, 1)$ indicating the proportion of points to be used for fitting the Generalized Pareto Distribution (GPD); a parameter $\alpha \in (0, 1)$ signifying the cut-off threshold for determining an anomaly; and a logical variable *unitize* indicating whether to scale the data. We briefly outline the steps below.

Algorithm 1 Lookout algorithm

Require: Data $\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$, proportion $\beta \in (0, 1)$, threshold $\alpha \in (0, 1)$, *unitize* $\in \{\text{True}, \text{False}\}$.

- 1: If *unitize*, scale the data to lie in the unit cube in $\mathbb{R}^m$.
 - 2: Compute the persistence homology barcode of the (possibly scaled) data cloud for dimension zero using the Vietoris-Rips diameter ([Ghrist 2008](#)).
 - 3: From the barcode, obtain the ordered death diameters $\{d_i\}_{i=1}^n$ for connected components, and their successive differences $\Delta d_i = d_{i+1} - d_i$.
 - 4: Let $d_* = d_{i_*}$ where $i_* = \arg \max \{\Delta d_i\}_{i=1}^n$; that is, d_* is the diameter corresponding to the lower point of the maximum Δd_i .
 - 5: Compute the kernel density estimates at the observations: $f_i = \hat{f}(\mathbf{y}_i)$, $i = 1, \dots, n$, where $\hat{f}$ is given by (4), $\mathbf{H} = h_n \mathbf{I}_m$, and $h_n = (d_*)^{2/m}$.
 - 6: Compute the leave-one-out kde values $f_{-i} = \frac{1}{n-1} (nf_i - \mathbf{H}^{-1/2} K(\mathbf{0}))$ for $i = 1, \dots, n$.
 - 7: Fit a Generalized Pareto Distribution (GPD) to the largest $1 - \beta$ of $\{-\log f_i\}_{i=1}^n$, obtaining estimates of the location, scale and shape parameters (μ, σ, ξ) .
 - 8: Calculate the probability of each observation by applying the estimated GDP to the leave-one-out kde values: $p_i = (1 - \beta)P(-\log f_{-i} \mid \hat{\mu}, \hat{\sigma}, \hat{\xi})$.
 - 9: If $p_i < \alpha$, declare $\mathbf{y}_i$ anomalous.
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The lookout algorithm provides density-based anomaly detection, where anomalies are identified as points with low density in the space of the data. A key feature of the algorithm is the use of minus log density values at the observations, known as “surprisals” in information theory ([Stone 2022](#)), and as “log scores” in statistics ([Good 1952](#)). If there are no anomalies in the data, the algorithm should identify α of the data as false positive “anomalies”.

While [Kandanaarachchi and Hyndman \(2022\)](#) demonstrated excellent results in applying their algorithm to a variety of simulated and real datasets, we see several potential areas for improvement that we aim to address in this paper. First, elements of the algorithm are not robust to outliers, including the min-max scaling, and the use of the largest gap in Rips death diameters. Second, the diagonal bandwidth matrix $\mathbf{H}$ does not account for correlations between variables, which can lead to suboptimal performance if the data have strong correlations. Third, it is not obvious that this bandwidth matrix choice will lead

to a consistent kernel density estimator, particularly for distributions with heavy tails. Finally, the GPD estimation made minimal assumptions about the distribution of the surprisal values, but we know that these are bounded above and below, and that fact can be used to improve the estimated GPD.

In the following section, we will explore the conditions under which the kernel density estimator is consistent for $h_n = d_*$, and the implications for the lookout algorithm. In particular, we will show that the conditions are not satisfied for distributions with heavy tails such as a power distribution, although it seems likely that they are satisfied for lighter-tailed distributions such as the Gaussian distribution. We will also show that an alternative choice of bandwidth, based on an upper quantile of the Rips death diameters, satisfies these conditions for a wide range of distributions.

The main mathematical approach for proving when a bandwidth is admissible uses the notion of crackle, introduced in [Adler et al. \(2014\)](#), which describes the persistent homology of noise. They proved asymptotic results for the persistent homology of the union of unit balls centered on iid samples of various distributions. There are qualitative differences between sets of samples taken from the Gaussian distribution (which do not “crackle”) compared to those from heavy-tailed distributions. We will adapt the methods to this setting where we wish to shrink the radius to 0, albeit slowly.

1.5 The modified lookout algorithm

As a result of these theoretical considerations, we propose three modifications to the lookout algorithm, described in detail in [Section 3](#).

1. Rather than use a max-min scaling of the data, we standardize the data using a robust covariance estimator. That is, we scale and rotate the data to ensure that the covariance matrix of the transformed data is approximately the identity matrix. This ensures that the data have zero pairwise correlations, which leads to more efficient kernel density estimates. It also leads to each variable having approximately unit variance, which provides a more robust scaling than min-max scaling.
2. Rather than using the lower point of the largest gap in Rips death diameters, we use an upper quantile of the Rips death diameters. This avoids the problem where anomalies affect the largest gap in the Rips diameters that cause a persistent homology death. It also ensures the consistency of the kernel density estimates under relatively weak conditions.
3. The estimated GPD shape parameter is constrained to be non-positive, which improves the stability of the GPD fitting. This is justified by noting that the kernel density estimates are bounded, so the limiting distribution under the Fisher-Tippett-Gnedenko theorem is a Weibull distribution, which has a negative shape parameter, ξ .

Note that we do not need to estimate the density of the underlying distribution directly. Instead, we can estimate the density of a transformation of the data, and identify anomalies in the transformed space. Hence, there is no need to undo the scaling or rotation.

1.6 Outline of the paper

The remainder of this paper is organized as follows. In [Section 2](#), we explore the theoretical properties of the lookout algorithm, particularly the choice of bandwidth based on Rips death diameters, and the consistency of the kernel density estimates produced by this choice. In [Section 3](#), we propose a modified lookout algorithm that improves performance and robustness under a broader range of conditions. We provide some empirical experiments, demonstrating the performance of the modified lookout algorithm on simulated and real datasets, first in contrast to the original lookout ([Section 4](#)), and then in contrast to other anomaly detection methods ([Section 5](#)). Finally, in [Section 6](#), we summarize our results and discuss future work.

2 Consistency and crackle

We will first prove a number of limit theorems about $d_{\omega(n)}$, that is the $\omega(n)$ -th death time (when ordered increasingly by size). For consistency of the kernel density estimator, we are interested in $d_{\omega(n)} \rightarrow 0$ and $nd_{\omega(n)} \rightarrow \infty$.

2.1 Limit of $nd_{\omega(n)}$

Theorem 4 *Let d_k denote the k th smallest death time of the 0-homology for the Rips filtrations of a set of n points sampled from probability distribution X over $\mathbb{R}^2$ such that the probability density function exists and is bounded above. Note that there are $n - 1$ finite death times.*

Choose any sequence of integers $\{\omega(n)\}$ such that $0 < \omega(n) < n$ for all n and $\lim_{n \rightarrow \infty} \omega(n) = \infty$. Then $nd_{\omega(n)} \rightarrow \infty$ as $n \rightarrow \infty$ with probability 1.

Proof For any function $r : \mathbb{N} \rightarrow (0, \infty)$, let $C(n, r)$ denote the number of components of $\cup_x B(x, r)$. Each death time corresponds to the reduction by one of the number of connected components. This implies that $nd_{\omega(n)} > r(n)$ occurs when $C(n, \frac{r(n)}{n}) > n - \omega(n)$.

For any bounded random variable X with range contained in $[0, N]$ we have

$$\mathbb{E}[X] \leq N(1 - \mathbb{P}(X \leq (N - \lambda))) + (N - \lambda)(\mathbb{P}(X \leq (N - \lambda))) = N - \lambda\mathbb{P}(X \leq N - \lambda)$$

and hence

$$\mathbb{P}(X \leq N - \lambda) \leq \frac{\mathbb{E}[X] - N}{\lambda}.$$

Since the number of components can only be an integer between 1 and n ,

$$\mathbb{P}\left(C\left(n, \frac{r(n)}{n}\right) \leq n - \omega(n)\right) \leq \frac{\mathbb{E}[C(n, \frac{r(n)}{n})] - n}{\omega(n)}.$$

It is thus sufficient to show that there exists a suitable sequence $r(n)$ such that $r(n) \rightarrow \infty$ as $n \rightarrow \infty$, and

$$\lim_{n \rightarrow \infty} \frac{\mathbb{E}[C(n, \frac{r(n)}{n})] - n}{\omega(n)} = 0.$$

For small radii r , a useful lower bound on the number of connected components is $n - E(n, r)$, where $E(n, r)$ is the number of edges in the graph connecting any two points are less than $2r$ apart, that is

$$\left| \mathbb{E}[C(n, \frac{r(n)}{n})] - n \right| \leq \mathbb{E}\left[E\left(n, \frac{r(n)}{n}\right)\right].$$

The expected number of edges is much easier to calculate than the expected number of components.

$$\mathbb{E}\left[E\left(n, \frac{r(n)}{n}\right)\right] = \frac{n(n-1)}{2} \mathbb{P}(d(x, y) < 2r \mid x, y \sim X) \leq \frac{n(n-1)}{2} K \pi \left(\frac{r(n)}{n}\right)^2 \leq \frac{K\pi}{2} r(n)^2$$

where K is the global upper bound on the probability density function of X . If we set $r(n) = \sqrt{2 \log(\omega(n))}$ (noting $\lim_{n \rightarrow \infty} r(n) = \infty$ as $\lim_{n \rightarrow \infty} \omega(n) = \infty$) we get

$$\lim_{n \rightarrow \infty} \frac{\left| \mathbb{E}\left[C\left(n, \frac{r(n)}{n}\right)\right] - n \right|}{\omega(n)} \leq \lim_{n \rightarrow \infty} \frac{K\pi \log(\omega(n))}{\omega(n)} = 0.$$

□

Theorem 5 *Let d_1 denote the smallest death time of the 0-homology for the Rips filtrations of a set of n points sampled from probability distribution over $\mathbb{R}^d$, $d \geq 3$, such that the density is bounded above.*

Then $nd_1 \rightarrow \infty$ as $n \rightarrow \infty$ with probability 1.

Proof Let $E(n, r)$ denote the number of edges in the graph which connects any two points that are less than $2r$ apart. If we set $r = \log(n)/n$ then

$$\mathbb{E}[E(n, r)] = \frac{n(n-1)}{2} \mathbb{P}(d(x, y) < 2 \log(n)/n \mid x, y \sim X) \leq \frac{n(n-1)}{2} K s_d \left(\frac{2 \log(n)}{n}\right)^d$$

where K is the global upper bound on the pdf of X and s_d is the volume of the unit ball in $\mathbb{R}^d$. As the number of edges is a non-negative integer, we have $\mathbb{P}(E(n, \log(n)/n) \neq 0) \leq \mathbb{E}[E(n, \log(n)/n)] \rightarrow 0$ as $n \rightarrow \infty$.

This implies that with high probability no edge appears in the Rips complex at radius $\log(n)/n$. As d_1 is the first time an edge appears, we have $\mathbb{P}(nd_1 > \log(n)) \rightarrow 1$ as $n \rightarrow \infty$. □

2.2 Limit of d_{n-1} for Gaussian Samples

In this section we will prove the complementary limit theorem needed for consistency results when we assume that the samples come from a Gaussian distribution. The proof showing that $d_{n-1} \rightarrow 0$ as $n \rightarrow \infty$ is a straightforward modification of the proof for Theorem 2.1 in [Adler et al. \(2014\)](#). The only real change is that we need to consider a sequence of decreasing radii for the balls, whereas in [Adler et al. \(2014\)](#)

the radii remain constant (value 1). We just need to ensure that the radii decrease slowly enough so that the union of balls remains connected. Due to this variation in the proof, we include the calculations for completeness.

Theorem 6 *Let S_n denote a random sample of n iid points from a fixed multivariate Gaussian in $\mathbb{R}^d$. Then for the Rips filtration over S_n we have $d_{n-1} \rightarrow 0$ as $n \rightarrow \infty$ almost surely.*

Proof We will first prove the result in the case of a standard Gaussian, where the probability density function is spherically symmetric and is $c \exp\{-\|x\|^2/2\}$ for some constant c . Denote $f(r) = c \exp\{-r^2/2\}$ for this density in terms of the radial coordinate.

Set $R_n := \sqrt{2(\log n + d \log \log n)}$ and $C_n = \sqrt{2(\log n - \log \log n)}$. R_n will be the radius outside which we will expect no samples to occur and C_n will represent the core of the point cloud - which we expect to be densely covered.

Cover $B(0, C_n)$ by a grid with side lengths g . Let $Y(n, C_n, g_n)$ be the event that at least one of the boxes in this grid is empty. Equation 3.1 in [Adler et al. \(2014\)](#) states:

$$\mathbb{P}(Y(n, C_n, g_n)) \leq (2g_n^{-1})^d (C_n)^d \exp\{-ng_n^d f(C_n)\}.$$

Set $g + n$ so $g_n^d = \frac{1}{c \log \log n}$. Further note that $f(C_n) = c \exp\{-\log n + \log \log n\} = c \frac{\log n}{n}$. Combined we get

$$\mathbb{P}(Y(n, C_n, g_n)) \leq 2^d c \log \log n (2(\log n - \log \log n))^{d/2} \exp\left(\frac{-nc \log n}{cn \log \log n}\right) \leq K \frac{(\log n)^{d/2} \log \log n}{\exp(\frac{\log n}{\log \log n})}$$

where K is a constant independent of n . For large n , $\log n > (d \log \log n) \log \log n$ and thus

$$\exp\left(\frac{\log n}{\log \log n}\right) > \exp(d \log \log n) = (\log n)^d.$$

Thus for large n we can bound

$$\mathbb{P}(Y(n, C_n, g_n)) \leq K \frac{\log \log n}{(\log n)^{d/2}}$$

which goes to 0 as $n \rightarrow \infty$.

Let $X(n, R_n)$ be the event that all the samples lie inside $B(0, R_n)$. From Theorem 4 in [Adler et al. \(2014\)](#) we have $\lim_{n \rightarrow \infty} \mathbb{E}[|S_n \cap B(0, R_n)|] = 0$. As $|S_n \cap B(0, R_n)|$ must be an integer between 0 and n this implies that $\lim_{n \rightarrow \infty} \mathbb{P}(X(n, R_n)) = 1$.

We want to bound the distance from any point in $B(0, R_n)$ to $B(0, C_n)$ which is the same as bounding $R_n - C_n$. We have $R_n + C_n > \sqrt{2(\log n + d \log \log n)} + \sqrt{2(\log n - \log \log n)} > 2\sqrt{\log n}$ so

$$\begin{aligned} R_n - C_n &= \frac{R_n^2 - C_n^2}{R_n + C_n} \\ &< \frac{2(\log n + d \log \log n) - 2(\log n - \log \log n)}{2\sqrt{\log n}} \\ &= \frac{(d+1) \log \log n}{\sqrt{\log n}} \end{aligned}$$

Suppose that both $Y(n, C_n, g_n)$ and $X(n, R_n)$ both hold. That is every sample lies in $B(0, R_n)$ and there is a sample in every box in the grid over C_n . Set

$$r(n) := \frac{2\sqrt{d}}{(c \log \log n)^{1/d}} + \frac{(d+1) \log \log n}{\sqrt{\log n}}$$

which was chosen so $r(n) > (R_n - C_n) + 2\sqrt{d}g_n$. By construction we have $\cup_{x \in S_n} B(x, r(n))$ must be connected.

In this case $d_{n-1} \leq r(n)$ as d_{n-1} is the supremum of radii r where $\cup_{x \in S_n} B(x, r)$ is disconnected.

As both $Y(n, C_n, g_n)$ and $X(n, R_n)$ occur with probability 1 as n grows to infinity we have $d_{n-1} \leq r(n) \rightarrow 0$ as $n \rightarrow \infty$ almost surely.

To then extend the result to general multivariate Gaussian, observe that our sample of points $\hat{S}_n$ is the linear transformation T of a set of points S_n of the standard Gaussian. If $\cup_{x \in S_n} B(x, r)$ is connected, then we can infer that $\cup_{x \in S_n} B(T(x), \|T\|r)$ is also connected. This implies that $d_{n-1}(\hat{S}_n) \leq \|T\|d_{n-1}(S_n)$ and thus $d_{n-1}(\hat{S}_n) \rightarrow 0$ as $n \rightarrow \infty$. $\square$

Corollary 7 *For point clouds sampled from multivariate Gaussian distributions, we have $d_* \rightarrow 0$ as $n \rightarrow \infty$.*

Proof Note that $i_* < n-1$ and so $d_* = d_{i_*} < d_{n-1}$ which we know limits to 0 almost surely from [Theorem 6](#). $\square$

Combining previous results implies that $\mathbf{H} = (d_*)^{2/m} \mathbf{I}_m$ results in a consistent density estimator of Gaussian samples in $\mathbb{R}^m$ for $m \geq 3$. From [Corollary 7](#) we have $d_* \leq d_{n-1} \rightarrow 0$ and from [Theorem 5](#) we have $nd_* \geq nd_1 \rightarrow \infty$. If $m = 2$, then to show that $\mathbf{H} = (d_*)^{2/m} \mathbf{I}_m$ results in a consistent density estimator, the only remaining task is to show $i(n) \rightarrow \infty$ as $n \rightarrow \infty$ almost surely. This seems intuitively clear, but technically challenging. It is not the focus of the new version of lookout, and thus we will leave this proof as an open problem.

2.3 A counterexample of a $d_{\omega(n)}$ which is not an admissible bandwidth

There are many potential indexing functions $\omega(n)$ that we can choose to decide which death time to use to compute the bandwidth matrix. We wish for this choice to determine a consistent density estimator.

This means that we need $d_{\omega(n)} \rightarrow 0$ as $n \rightarrow \infty$.

If our samples are Gaussian, then we know that $d_{\omega(n)}$ leads to a consistent kernel density estimator whenever $\lim_{n \rightarrow \infty} \omega(n) = \infty$. However, when samples come from a fat-tailed distribution, we can construct examples where $d_{\omega(n)}$ does not converge to 0 almost surely, which implies that $d_{\omega(n)}$ does not give a consistent kernel density estimator.

Consider the distribution on $\mathbb{R}^d$ with the density function $f(x) = \frac{c_d}{1+\|x\|^{d+1}}$ where c_d is the appropriate constant. For each radius R the probability a sample has modulus at least R (that is it lies outside $B(0, R)$) is

$$p_R = \int_R^\infty \frac{c_d s_{d-1} r^{d-1}}{1+r^{d+1}} dr$$

where s_{d-1} is the area of the $(d-1)$ -sphere. We have $p_R \in (\frac{c_d s_{d-1}}{2R}, \frac{c_d s_{d-1}}{R})$ for large enough R .

Set $R_n = \frac{2n^{2/d}}{c_d s_{d-1}}$. Let S_n be the number of samples that lie outside $B(0, R_n)$. We have $S_n \sim \mathcal{B}(n, p_{R_n})$ and with $p_{[R_n, \infty)} \in (n^{-2/d}, 2n^{-2/d})$. That implies that there is a $\delta_1 > 0$ independent of n such that

$$\mathbb{P}(S_n \in (n^{1-2/d}, 2n^{1-2/d})) > \delta_1.$$

Given a sample lies in $B(0, R)$, the probability that a sample has modulus in $[R-1, R)$ (that is it lies in $B(0, R_n) \setminus B(0, R_n - 1)$) is

$$p_{[R-1, R)} = \frac{\int_{R-1}^R \frac{c_d s_{d-1} r^{d-1}}{1+r^{d+1}} dr}{1 - p_R} < \frac{\frac{c_d s_{d-1}}{R} - \frac{c_d s_{d-1}}{R-1}}{1 - \frac{c_d s_{d-1}}{R}} < \frac{2c_d s_{d-1}}{R^2}$$

for sufficiently large R . For $R_n = \frac{2n^{2/d}}{c_d s_{d-1}}$ we have $p_{[R-1, R)} \leq n^{-2/d}$ for large n .

Let $\hat{S}_n$ the number of samples in $B(0, R_n) \setminus B(0, R_n - 1)$ conditioned on there being S_n samples outside $B(0, R_n)$. We have $\hat{S}_n \sim \mathcal{B}(n - S_n, p_{[R_n, R_n - 1)})$. We can couple $\mathcal{B}(n - S_n, p_{[R_n, R_n - 1)})$ to $\mathcal{B}(n, p_{[R_n, R_n - 1)})$ to imply that $\mathbb{P}(\hat{S}_n \leq n^{1-2/d}) \geq \frac{1}{2}$.

Consider the Rips complex at radius 1. We want to lower bound the probability that all the points outside $B(0, R)$ are isolated. A sufficient condition for this is that there are no edges between points outside $B(0, R-1)$.

$$\mathbb{P}(d(x, y) \leq 1 \mid \|x\|, \|y\| \geq R-1) = \frac{1}{\mathbb{P}(\|x\|, \|y\| \geq R-1)} \int_{R-1}^{\infty} \frac{c_d s_{d-1} r^{d-1}}{1+r^{d+1}} \mathbb{P}(y \in B(x, 1) \mid |x| = r) dr$$

If $|x| = r$ (r sufficiently large), then the probability that another sample y lies in $B(x, 1)$ is bounded above by C/r^d for some constant C independent of r . This implies that there exists a constant C_1 independent of R such that

$$\mathbb{P}(d(x, y) \leq 1 \mid \|x\|, \|y\| > R-1) < (R-1)^2 C_1 \int_{R-1}^{\infty} \frac{1}{r^{d+2}} dr \leq \frac{C_2}{R^{d-1}}.$$

for some constant C_2 independent of R (for sufficiently large R).

Let Y_n be the number of edges in the Rips complex at radius 1 between points outside $B(0, R_n - 1)$.

$$\mathbb{E}[Y_n \mid S_n \leq 2n^{1-2/d} \text{ and } \hat{S}_n \leq n^{1-2/d}] \leq \frac{(3n^{1-2/d})^2}{2} C_2 (n^{2/d})^{-d+1} = \frac{3C_2}{2n^{2/d}}$$

For any $\delta_2 \in (0, 1)$ we have $\mathbb{E}[Y_n \mid S_n < 2n^{1-2/d} \text{ and } \hat{S}_n \leq n^{1-2/d}] < 1 - \delta_2$ for sufficiently large n . When $\mathbb{E}[Y_n \mid S_n < 2n^{1-2/d}] < 1 - \delta_2$ we have $\mathbb{P}(Y_n = 0 \mid S_n < 2n^{1-2/d} \text{ and } \hat{S}_n \leq n^{1-2/d}) > \delta_2$.

Combined we have

$$\mathbb{P}(S_n \in (n^{1-2/d}, 2n^{1-2/d}) \text{ and } \hat{S}_n \leq n^{1-2/d} \text{ and } Y_n = 0) > \frac{1}{2}\delta_1\delta_2$$

for sufficiently large n . When $S_n \in (n^{1-2/d}, 2n^{1-2/d})$ and $\hat{S}_n \leq n^{1-2/d}$ and $Y_n = 0$ we are guaranteed that at least $n^{1-2/d}$ points are isolated in the Rips complex at radius 1. This implies that there are at least $n^{1-2/d}$ connected components and $d_{n-n^{1-2/d}} \geq 1$.

If we set $\omega(n) = n - n^{1-2/d}$, then we do *not* have the limit $d_{\omega(n)} \rightarrow 0$ as $n \rightarrow \infty$ with probability 1 which means that $d_{\omega(n)}$ does not give a consistent estimator.

2.4 Quantile death times $d_{\alpha(n-1)}$ are an admissible bandwidth

Despite the counterexample in the previous subsection, we can find consistent estimators using sequences of death times under very mild assumptions of the distribution (even with fat tails!). An approach that generally works is to a percentile of the death times. That is $d_{\alpha(n-1)}$ for some $\alpha \in [0, 1]$. Recall that there are only $n - 1$ finite death times, so d_{n-1} is the largest finite death value. Strictly speaking, we will need an integer value, so you can interpret $\alpha(n - 1)$ as either the floor or the ceiling; the same limits will hold in either case.

The following is a standard useful lemma, but we include it for completeness.

Lemma 8 *Let $U_1, \dots, U_l \subset \mathbb{R}^d$ and X a random sample such that $\mathbb{P}(X \in U_i) > \delta$ for all i . Let $\{x_1, \dots, x_n\}$ be i.i.d. draws of X . If Z_i is the event that at no samples x_j lie in U_i then*

$$\mathbb{P}(\cup Z_i) \leq le^{-n\delta}.$$

Proof For a single draw x_j we have $\mathbb{P}(x_j \notin Z_i) \leq 1 - \delta \leq e^{-\delta}$ and thus for n draws $\mathbb{P}(Z_i) \leq e^{-n\delta}$. Combining over all the Z_i we gain the upper bound $\mathbb{P}(\cup Z_i) \leq \sum_i \mathbb{P}(Z_i) \leq le^{-n\delta}$. $\square$

It is useful to have notation for offsets of Euclidean subsets. For any subset $L \subset \mathbb{R}^d$ and any $\delta > 0$, let L^δ denote the set $\{y + v \mid y \in L, \|v\| \leq \delta\}$.

Lemma 9 *Let $f : \mathbb{R}^d \rightarrow [0, \infty)$ be a Lipschitz probability density function (with Lipschitz constant M), and $L = f^{-1}[c, \infty)$ be a non-empty superlevel set of f for some $c > 0$. For all $\epsilon \in (0, c/2M]$ there exists $\{x_1, \dots, x_l\} \subset L$ such that $\{B(x_i, \epsilon)\}$ is a cover of L , $l \leq \frac{2^{d+2}}{3cb_d\epsilon^d}$, and $\mathbb{P}(X \in B(x_i, \epsilon)) > \frac{cb_d}{2}\epsilon^d$ for all i . Here b_d is the volume of the unit ball in $\mathbb{R}^d$.*

Proof Take a maximal $(\epsilon/2)$ -separated subset $S = \{x_1, x_2, \dots, x_l\} \subseteq L$. This means that $|x_i - x_j| \geq \epsilon$ for all $i \neq j$ and that no additional point in L can be added to S without breaking this condition. By construction this

implies that $L \subset \cup_i B(x_i, \epsilon)$ as otherwise any missed location in L could be added as an additional point and the set remain $(\epsilon/2)$ -separated.

As each $B(x_i, \epsilon/2) \subset L^{\epsilon/2}$ and f is Lipschitz with Lipschitz constant M , we have $L^{\epsilon/2} \subseteq f^{-1}[c - M\epsilon/2, \infty)$. For $\epsilon < c/2M$ we have $B(x_i, \epsilon/2) \subset f^{-1}[3c/4, \infty)$ so $\mathbb{P}(X \in B(x_i, \epsilon/2)) \geq \frac{3cb_d\epsilon^d}{2^{d+2}}$. Each of the $B(x_i, \epsilon/2)$ are disjoint so $l \frac{3cb_d\epsilon^d}{2^{d+2}} \leq \mathbb{P}(X \in \cup_i B(x_i, \epsilon/2)) \leq 1$. We thus get $l \leq \frac{2^{d+2}}{3cb_d\epsilon^d}$.

As $B(x_i, \epsilon) \subset f^{-1}[c - M\epsilon, \infty) \subset f^{-1}[c/2, \infty)$ we have $\mathbb{P}(X \in B(x_i, \epsilon)) \geq \frac{cb_d\epsilon^d}{2}$. $\square$

Theorem 10 *Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$ be a Lipschitz probability density function (with Lipschitz constant M) for X . For any $\alpha \in (0, 1)$, the limit $d_{\alpha(n-1)} \rightarrow 0$ converges almost surely, where $d_{\alpha(n-1)}$ is the α quantile of finite death times.*

Proof Fix $\alpha \in [0, 1)$. We have $d_{\alpha(n-1)} < r_n$ whenever $\cup_i B(x_i, r_n)$ has less than $n(1 - \alpha)$ connected components.

Fix $c_\alpha > 0$ such that for $L = \{x | f(x) \geq c_\alpha\}$ we have $\mathbb{P}(X \notin L_\alpha) < (1 - \alpha)/2$. Choose a sequence $\epsilon_n := \left(\frac{2 \log n}{c_\alpha b_d n}\right)^{1/d}$ so that

$$\frac{c_\alpha b_d \epsilon_n^d}{2} = \frac{\log n}{n}.$$

(where b_d is the volume of the unit ball in $\mathbb{R}^d$). Note that for $\epsilon_n \in (0, c/2M)$ for large enough n .

By Theorem 9 there exists a cover of L by l_n balls $\{B(y_i, \epsilon_n)\}$, with

$$l_n \leq \frac{2^{d+1}}{c_\alpha b_d \epsilon_n^d} = \frac{2^d n}{\log n}$$

and $\mathbb{P}(X \in B(y_i, \epsilon_n)) > \frac{cb_d \epsilon_n^d}{2} = \frac{\log n}{n}$.

If Z_i is the event that there are no samples in $B(y_i, \epsilon)$. By Theorem 8

$$\mathbb{P}(\cup Z_i) \leq \frac{2^d n}{\log n} e^{-n \frac{\log n}{n}} = \frac{2^d}{\log n}.$$

This implies that as $n \rightarrow \infty$ we have at least one sample in each of the $B(y_i, \epsilon_n)$ almost surely. All points with the same ball $B(y_i, \epsilon_n)$ will be connected in the Rips complex with radius $2\epsilon_n$. This implies that the number of connected components when restricting to points inside L_α will be bounded above by $l_n \leq \frac{2^d n}{\log n}$ (almost surely).

Let $Y(L_\alpha, n)$ denote the random variable for the number of our n sample points that are not in L_α . We want

$$Y(L_\alpha, n) + \frac{2^d n}{\log n} < n(1 - \alpha)$$

for large n . As $\frac{2^d n}{\log n} < \frac{n(1-\alpha)}{6}$ for sufficiently large n , it is sufficient to show $Y(L_\alpha, n) < 5n(1 - \alpha)/6$ for large n .

From the construction of L_α we have $Y(L_\alpha, n)$ is a binomial distribution with mean np and variance $np(1 - p)$, with $p < (1 - \alpha)/2$. We can use Chebyshev's inequality to bound $\mathbb{P}(Y(L_\alpha, n) > 5n(1 - \alpha)/6)$.

$$\begin{aligned} \mathbb{P}(Y(L_\alpha, n) > 5n(1 - \alpha)/6) &\leq \mathbb{P}(|Y(L_\alpha, n) - np| > 5n(1 - \alpha)/6 - n(1 - \alpha)/2) \\ &= \mathbb{P}(|Y(L_\alpha, n) - np| > n(1 - \alpha)/3) \\ &\leq \frac{\text{Var}(Y)}{(n(1 - \alpha)/3)^2} \\ &\leq \frac{9np(1 - p)}{n^2(1 - \alpha)^2} \end{aligned}$$

$$\leq \frac{9}{2n(1-\alpha)}$$

This clearly goes to zero as n goes to infinity.

If there are less than $n(1-\alpha)$ connected components in $\cup_j B(x_j, \epsilon_n)$ then $d_{\alpha(n-1)} < \epsilon_n$. Thus we have proved $\mathbb{P}(d_{\alpha(n-1)} > \epsilon_n) \rightarrow 0$ as $n \rightarrow \infty$. As $\epsilon_n \rightarrow 0$ as $n \rightarrow \infty$ we have $d_{\alpha(n-1)} \rightarrow 0$ almost surely. $\square$

We can combine Theorem 10 with Theorem 4 (for distributions over $\mathbb{R}^2$) or Theorem 5 (for distributions over $\mathbb{R}^d$ with $d > 2$) to show $d_{\alpha(n-1)}$ is an admissible bandwidth as a corollary.

Corollary 11 *Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a Lipschitz probability density function for X . Fix $\alpha \in (0, 1)$. Then $d_{\alpha(n-1)} \rightarrow 0$ (where $d_{\alpha(n-1)}$ is the α quantile of finite death times) is an admissible bandwidth.*

3 Lookout algorithm v2

In light of the preceding theoretical considerations, we propose three modifications to the lookout algorithm, described in this section.

3.1 Standardization

Let $\mathbf{Y}$ be the $m \times n$ data matrix with rows $\mathbf{y}_1, \dots, \mathbf{y}_n$. Then the *orthogonalized Gnanadesikan-Kettenring estimator* provides an iterative robust estimate of Σ , the covariance matrix of $\mathbf{Y}$ (Maronna and Zamar 2002). Let

$$\hat{\Sigma}^{-1} = \mathbf{U}' \mathbf{\Lambda} \mathbf{U} \tag{5}$$

denote its eigendecomposition. Then, the data can be scaled by $\mathbf{U}$, giving $\mathbf{Z} = \mathbf{U}\mathbf{Y}$, so that the covariance matrix of the scaled data $\mathbf{Z}$ is (approximately) the identity matrix. This can be thought of as both scaling and rotating the data; it is the multivariate equivalent of computing robust z-scores.

We prefer this to the min-max scaling used in the original lookout algorithm, as it is robust to outliers and removes the correlations between variables, leading to more efficient kernel density estimates.

3.2 Bandwidth matrix selection

The Rips death diameters are computed from the scaled and rotated data $\mathbf{Z}$. Let $d_1, \dots, d_n$ denote the (ordered) Rips death diameters computed from the sample $\mathbf{z}_1, \dots, \mathbf{z}_n$, and let $d^* = d_{0.9(n-1)}$ denote the 0.9 sample quantile computed from $\{d_1, \dots, d_n\}$ (Hyndman and Fan 1996).

A kernel density estimator is applied to the scaled data $\{\mathbf{z}_1, \dots, \mathbf{z}_n\}$ using the bandwidth matrix $\mathbf{H} = (d^*)^{2/m} \mathbf{I}$, where m is the dimension of the data. Under this estimator, up to 10% of the data may be anomalous without affecting the bandwidth matrix selection, and the density estimator is consistent under Corollary 11.

Further, the kernels are now aligned with the data, as both have covariance matrices that are scalar multiples of the identity matrix. While alignment is not necessary for consistency, it does improve the efficiency of the kernel density estimates (Chacón and Duong 2018).

3.3 Bounds for kernel density estimates

In the following lemma we show that the kernel density estimates at the observations, and their negative logarithms, are bounded by constants that depend only on the dimension m and the sample size n .

Lemma 12 *Let $\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$ be n data points in $\mathbb{R}^m$, with kernel density estimate given by*

$$\hat{f}(\mathbf{y}) = \frac{1}{n} \sum_{i=1}^n |\mathbf{H}|^{-1/2} K\left(\mathbf{H}^{-1/2}(\mathbf{y} - \mathbf{y}_i)\right),$$

where $K(\mathbf{u})$ is a multivariate kernel function that is bounded below by 0 and above by $K(\mathbf{0}) = M$, and $\mathbf{H}$ is an $m \times m$ symmetric positive-definite bandwidth matrix. Then for all $j \in \{1, \dots, n\}$, the kernel density estimates at the observations, $\hat{f}(\mathbf{y}_j)$, and their negative logarithms, $-\log \hat{f}(\mathbf{y}_j)$, are bounded by constants that depend only on m and n .

Proof

$$\frac{M|\mathbf{H}|^{-1/2}}{n} \leq \hat{f}(\mathbf{y}_j) = \frac{1}{n} \sum_{i=1}^n |\mathbf{H}|^{-1/2} K\left(\mathbf{H}^{-1/2}(\mathbf{y}_j - \mathbf{y}_i)\right) \leq M|\mathbf{H}|^{-1/2}.$$

The constant M may depend on m . □

For example, if we use the multivariate Gaussian kernel $K(\mathbf{u}) = \frac{1}{(2\pi)^{m/2}} \exp\left(-\frac{1}{2}\|\mathbf{u}\|^2\right)$, then $M = \frac{1}{(2\pi)^{m/2}}$. If we use the multivariate Epanechnikov kernel $K(\mathbf{u}) = \frac{m+2}{2c_m} (1 - \|\mathbf{u}\|)_+$, where c_m is the volume of the unit sphere in $\mathbb{R}^m$, and $u_+ = \max(0, u)$, then $M = (m+2)/(2c_m)$.

Under Theorem 1, the limiting distribution of the extremes of $-\log \hat{f}(\mathbf{y}_j)$ is a Weibull distribution, corresponding to a Generalized Extreme Value distribution with $\xi < 0$. Thus, when we fit a Generalized Pareto Distribution (GPD) to the largest $1 - \beta$ of $\{-\log \hat{f}(\mathbf{y}_j)\}_{j=1}^n$, we can constrain the shape parameter to be non-positive, which improves the stability of the GPD fitting.

3.4 Modified lookout algorithm

We are now in a position to describe the modified lookout algorithm.

As with the original lookout algorithm, the values of p_i can also act as anomaly scores, with lower values indicating more anomalous points.

The default values for the parameters are $\alpha = 0.001$, $\beta = 0.90$ and $\gamma = 0.97$. The default values are used in all experiments reported here, unless explicitly stated otherwise. Scaling is also applied by default, unless otherwise stated.

Algorithm 2 Modified lookout algorithm

Require: Data $\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$, proportion $\beta \in (0, 1)$, threshold $\alpha \in (0, 1)$, and quantile level $\gamma \in (0, 1)$.

- 1: Let $\mathbf{Y}$ be the $m \times n$ data matrix with rows $\mathbf{y}_1, \dots, \mathbf{y}_n$.
 - 2: Optional scaling: Calculate the eigendecomposition (5), and rotate and scale the data using $\mathbf{U}$, giving $\mathbf{Z} = \mathbf{U}\mathbf{Y}$, with scaled observations $\mathbf{z}_1, \dots, \mathbf{z}_n$ comprising the rows of $\mathbf{Z}$.
 - 3: Compute the persistence homology barcode of the data cloud $\mathbf{Z}$ (set $\mathbf{Z} = \mathbf{Y}$ if no scaling is applied) for dimension zero using the Vietoris-Rips diameter (Ghrist 2008), and obtain the ordered death diameters $\{d_i\}_{i=1}^n$ for connected components.
 - 4: Let $d^* =$ the γ sample quantile computed from $\{d_i\}_{i=1}^n$.
 - 5: Compute the kernel density estimates at the observations: $f_i = \hat{f}(\mathbf{y}_i)$, $i = 1, \dots, n$, where $\hat{f}$ is given by (4), $\mathbf{H} = h_n \mathbf{I}_m$, and $h_n = (d^*)^{2/m}$.
 - 6: Compute the leave-one-out kde values $f_{-i} = \frac{1}{n-1} (nf_i - \mathbf{H}^{-1/2} K(\mathbf{0}))$ for $i = 1, \dots, n$.
 - 7: Fit a Generalized Pareto Distribution (GPD) to the largest $1 - \beta$ of $\{-\log f_i\}_{i=1}^n$, constraining the shape parameter to be non-positive, obtaining estimates of the location, scale and shape parameters (μ, σ, ξ) .
 - 8: Calculate the probability of each observation by applying the estimated GDP to the leave-one-out kde values: $p_i = (1 - \beta)P(-\log f_{-i} \mid \hat{\mu}, \hat{\sigma}, \hat{\xi})$.
 - 9: If $p_i < \alpha$, declare $\mathbf{y}_i$ anomalous.
-

4 Comparing new lookout with the older version

Next we compare the old and the new versions of lookout. We conduct four experiments that explore different aspects of the updated algorithm, and illustrate the output using a set of showcase examples. All experiments have anomalies and non-anomalies. Experiments 1 and 2 allow the anomalies to move away from the non-anomalous distribution. These two experiments compare the sensitivity of the old and new versions of lookout to identifying anomalies as they become more obvious. Experiments 3 and 4 focus on having anomalies on the boundary of the non-anomalous region. For these two experiments we increase the number of data points and explore the performance of the two algorithm versions for different sample sizes.

4.1 Experiment 1: Different anomaly rates for Gamma distribution

For experiment 1 we consider the $\text{Gamma}(a, r)$ distribution with shape a and rate r and generate data points with different rates r . We select $\text{Gamma}(2, 2)$ as the non-anomalous distribution and $\text{Gamma}(2, r)$ as the anomalous distribution where $r \in \{0.1, 0.2, \dots, 1\}$. Let $Y_i = (Y_{i1}, Y_{i2})$ where Y_{i1} and Y_{i2} are independent with $Y_{ij} \sim \text{Gamma}(2, 2)$ for the non-anomalous distribution and $Y_{ij} \sim \text{Gamma}(2, r)$ for the anomalous distributions. We consider 10 replications of the experiment for each value of r . For each replication we generate 500 non-anomalies and 10 anomalies.

Fig. 1 shows the data and results of Experiment 1. As the rate r of the anomalous Gamma distribution increases, the anomalies and non-anomalies become more similar. Thus, we expect a decrease in the true positive rate and an increase in the false negative rate as r increases. The anomalies are maximally different from non-anomalies when $r = 0.1$. We see that new lookout performs better on average than the old lookout algorithm across all values of r , with the largest differences for the smallest values of r .

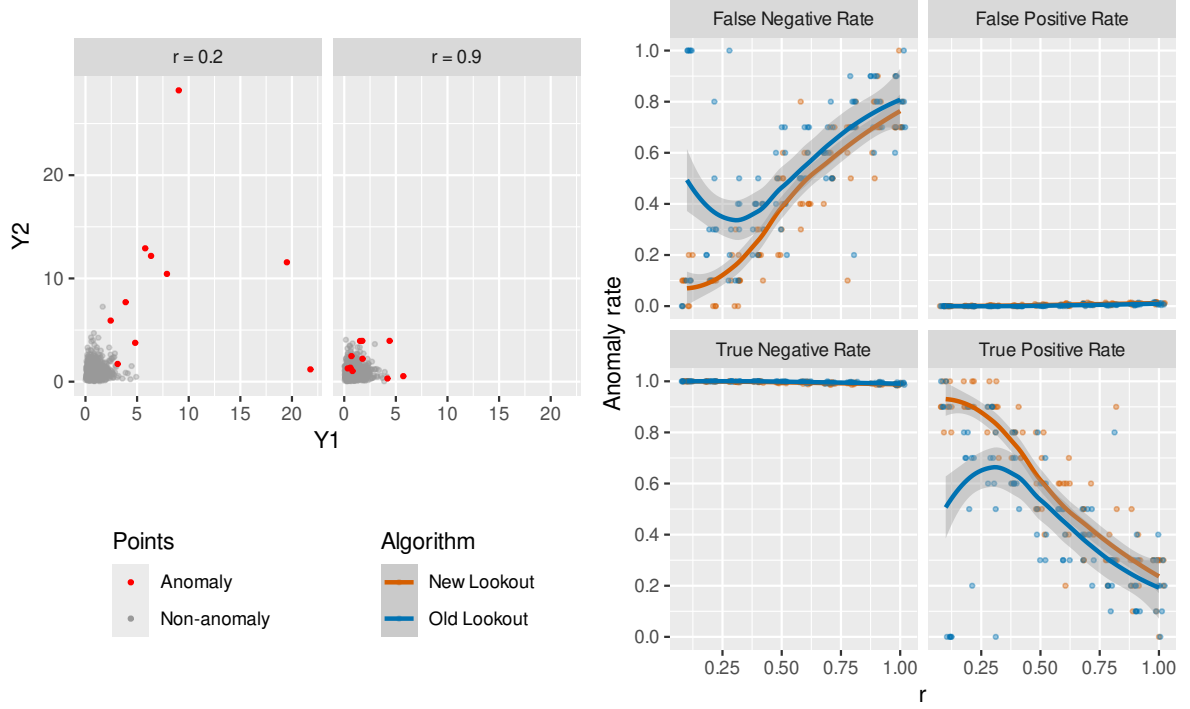


Fig. 1: Experiment 1 data and results. Left: data for two iterations of the experiment with $r = 0.2$ and $r = 0.9$. Anomalies with lower values of r are easier to identify. Right: the relative performance of the two algorithms. The points in the right panel are slightly jittered horizontally to avoid overlapping points.

4.2 Experiment 2: Different anomaly means for Normal distribution

For experiment 2 we consider the normal distribution $\mathcal{N}(\mu, \sigma)$ with mean μ and standard deviation σ . We let $Y_i = (Y_{i1}, Y_{i2})$ where Y_{i1} and Y_{i2} are independent with $Y_{ij} \sim \mathcal{N}(\mu, \sigma)$. The non-anomalous distribution is $\mathcal{N}(0, 1)$. The anomalous distribution has $\mu \in \{2.50, 2.75, \dots, 3.75, 4.00\}$ and $\sigma = 0.2$. We consider 1000 non-anomalies and 10 anomalies, and we produce 10 replicates for each value of μ .

Fig. 2 shows Experiment 2 data and results. As μ increases, the anomalies shift away from the non-anomalies. Thus, we expect the true positive rate to increase and the false negative rate to decrease with μ . Here, the two versions of the lookout algorithm are statistically indistinguishable, across all values of μ .

4.3 Experiment 3: As N increases - Normal distribution

It is often difficult to distinguish anomalies from legitimate observations because they can appear similar. In such cases, anomalies are not well separated from the non-anomalous points. In this experiment, we explore the efficacy of *lookout* when anomalies are close to regions containing non-anomalous points.

We consider n non-anomalies for $n \in \{1000, 2000, \dots, 10000\}$ and $0.005 \times n$ anomalies. We let $Y_i = (Y_{i1}, Y_{i2})$ where Y_{i1} and Y_{i2} are independent, with $Y_{ij} \sim \mathcal{N}(\mu, \sigma)$. For the non-anomalies, we set $\mu = 0$ and $\sigma = 1$. We generate anomalies in a ring just beyond most of the non-anomalous points, by first

sampling $\mu_{i1} \sim \mathcal{U}(-\sqrt{2} \times 2.2, \sqrt{2} \times 2.2)$ and letting $\mu_{i2} = \sqrt{2 \times 2.2^2 - \mu_{i1}^2}$. Then we sample the anomalies (Y_{i1}, Y_{i2}) where $Y_{i1} \sim \mathcal{N}(\mu_{i1}, 0.1)$ and $Y_{i2} \sim \mathcal{N}(\mu_{i2}, 0.1)$. We produce 10 replications for each value of n .

Fig. 3 shows the data of a single run and the results of experiment 3. Both new and old lookout give low false positive rates and high true negative rates. However, we see that new lookout has a larger true positive rate and a lower false negative rate compared to old lookout, with statistically significant differences for smaller sample sizes.

4.4 Experiment 4: As N increases - Gamma distribution

This experiment is similar to experiment 3, but we set $Y_{ij} \sim \text{Gamma}(2, 2)$ for the non-anomalous points. To obtain anomalous points, we first generate n points $Y_i = (Y_{i1}, Y_{i2})$ where Y_{i1} and Y_{i2} are independent, with $Y_{ij} \sim \text{Gamma}(2.2, 2)$. Then we consider the subset $\{Y_i : \|Y_i\| > \|X\|_{0.99}\}$ where $\|Y_i\| = \sqrt{Y_{i1}^2 + Y_{i2}^2}$ denotes the distance from the origin to Y_i and $\|X\|_{0.99}$ denotes the 99th percentile of $\left\{\sqrt{Y_{i1}^2 + Y_{i2}^2}\right\}_{i=1}^n$. From this subset we sample $0.005 \times n$ points and label them as anomalies. Each iteration for a given n is repeated 10 times. Fig. 4 shows the data from an iteration of the experiment with $n = 10000$ and the results for the new and old version of *lookout*. The two versions of lookout are not significantly different, other than at the largest sample sizes, where the older version does slightly better.

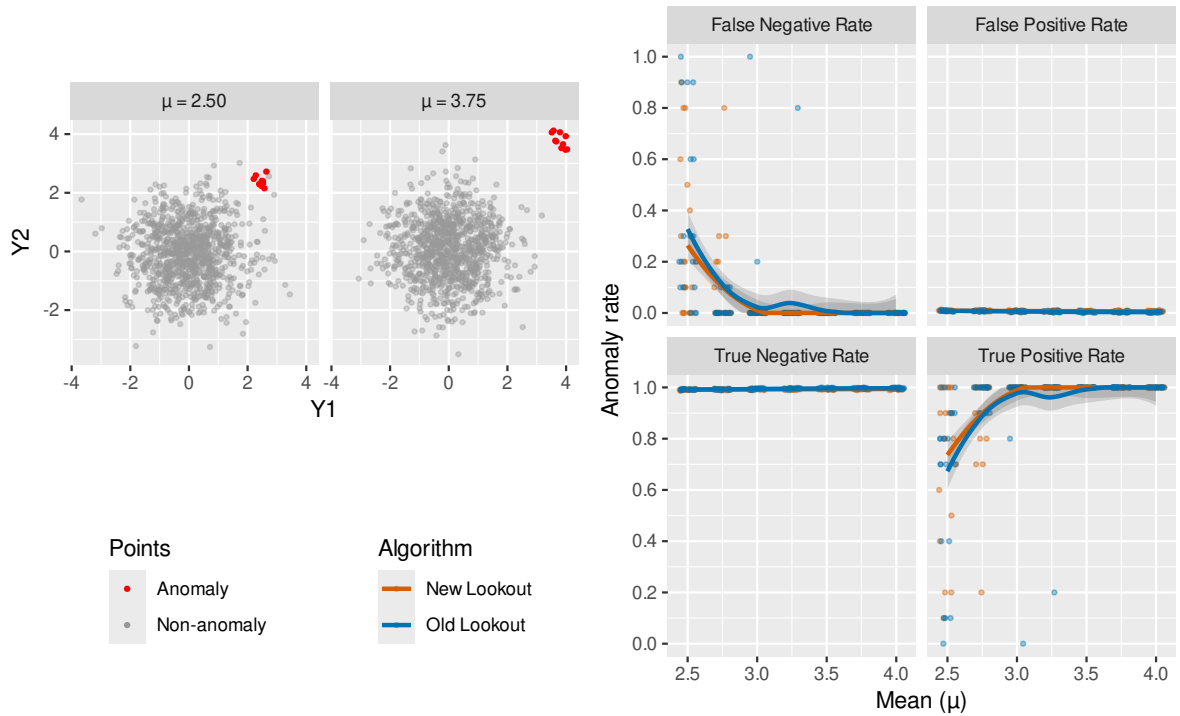


Fig. 2: Experiment 2 data and results. Left: data for two iterations of the experiment with $\mu = 2.5$ and $\mu = 3.75$. Anomalies with higher values of μ are easier to identify. Right: the relative performance of the two algorithms. The points in the right panel are slightly jittered horizontally to avoid overlapping points.

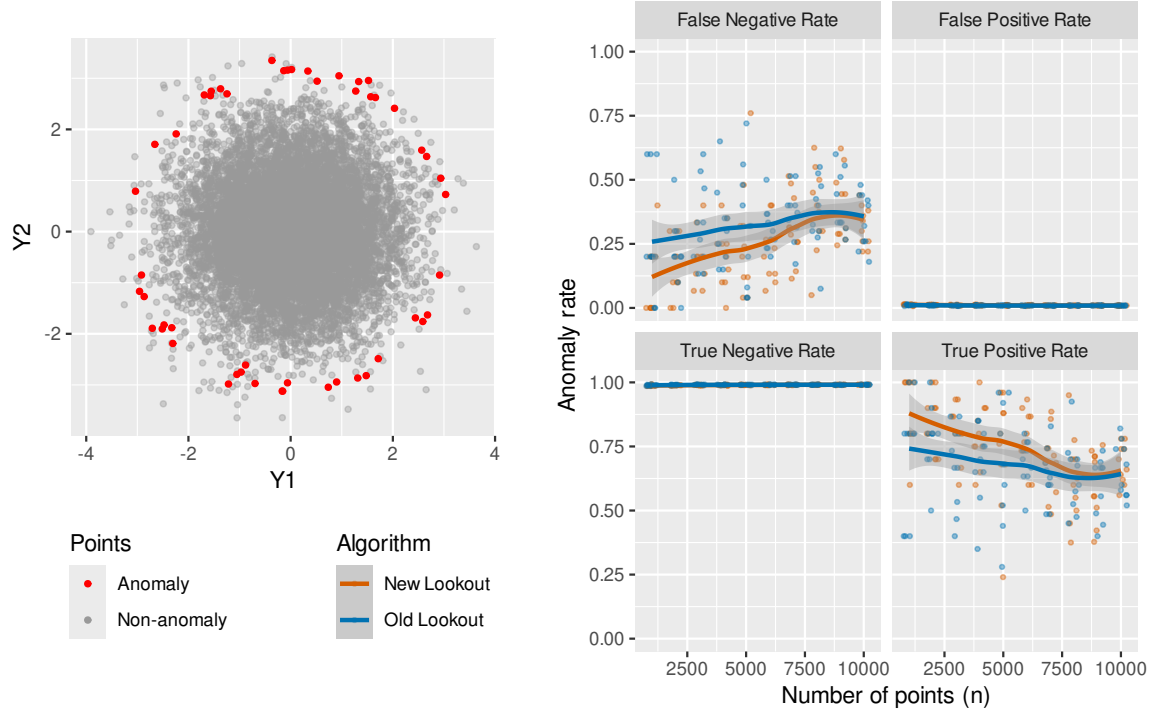


Fig. 3: Experiment 3 data and results using Normal distributions. Left: data for $n = 10000$. Right: the relative performance of the two algorithms. The points in the right panel are slightly jittered horizontally to avoid overlapping points.

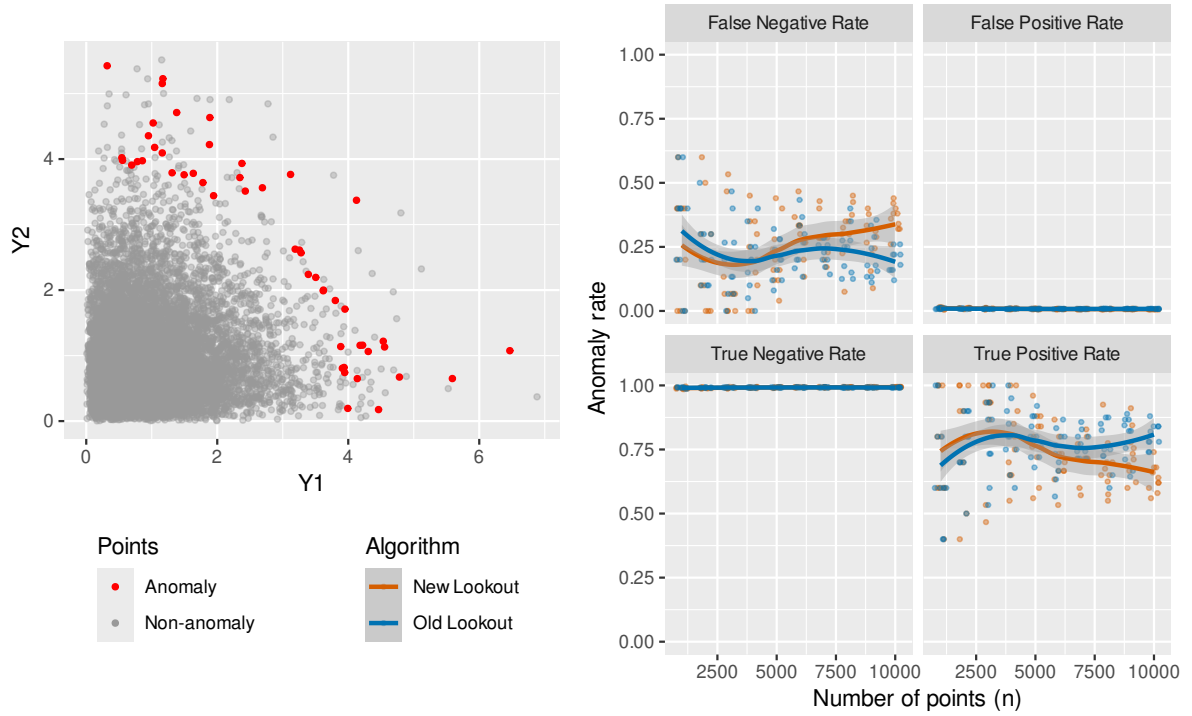


Fig. 4: Experiment 4 data and results using Gamma distributions. Left: data for $n = 10000$. Right: the relative performance of the two algorithms. The points in the right panel are slightly jittered horizontally to avoid overlapping points.

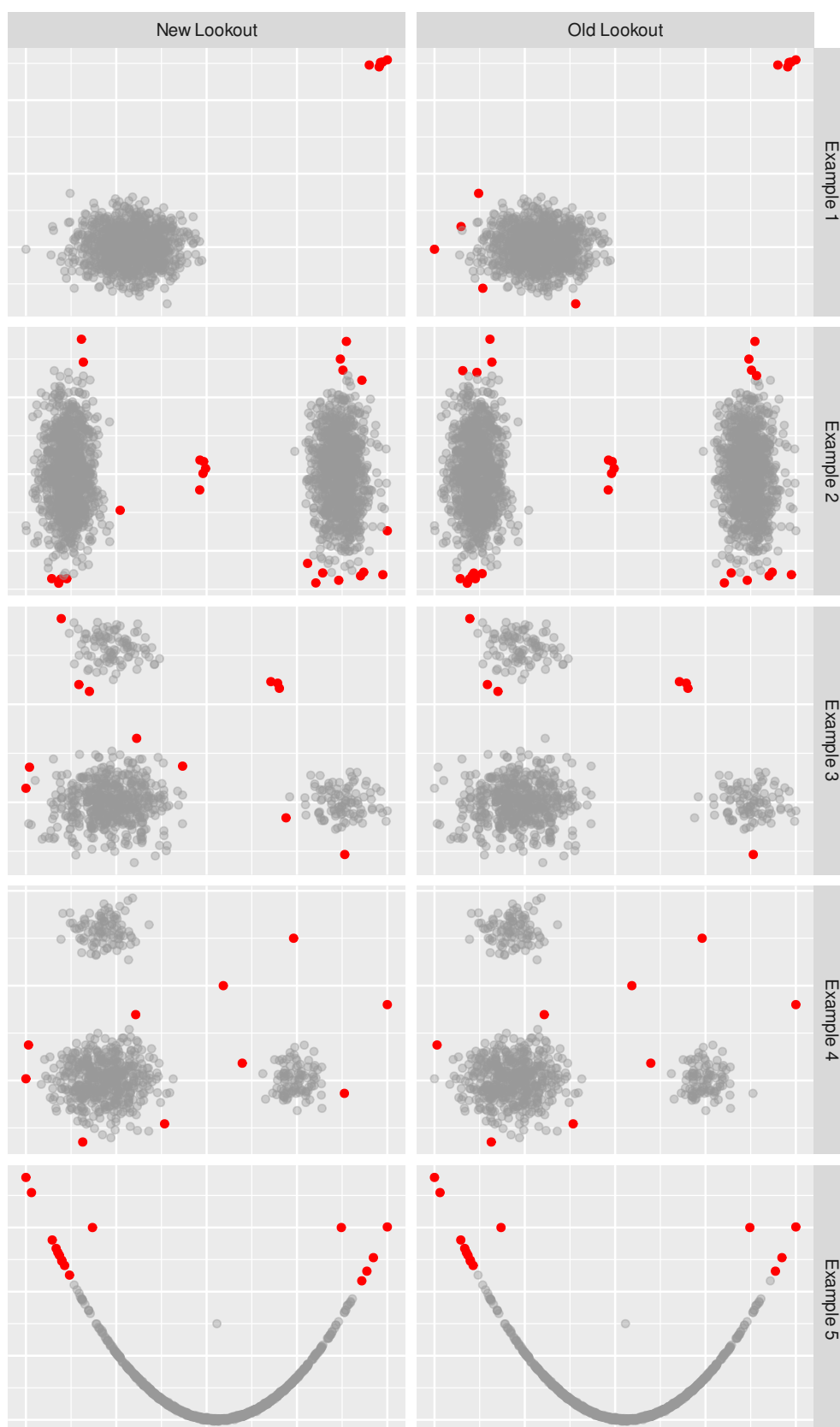


Fig. 5: Comparing new lookout with the older version on some showcase examples

4.5 Showcase examples

Fig. 5 shows some examples of new and old versions of lookout with identified anomalies in red. The purpose of this exercise is to check if the new version agrees with the old version on obvious anomalies. The graphs on the first row have anomalies at the top right-hand corner, which are identified by both algorithms, but the old algorithm also finds some false positives in the lower cloud of points. The second row depicts a bi-modal distribution with anomalies placed in the middle. Both versions of lookout identify some non-anomalous points as anomalies, with the new version giving fewer false positives. The third row has 3 clusters of normally distributed points with anomalies placed away from the three clusters. Here, the older version of the algorithm finds four false positives compared to nine with the new algorithm. The same setting is repeated in the fourth row with the anomalies spread out, and the two algorithms are almost identical. The last row has non-anomalous points in a parabola and the anomalous points outside of it. This is a tricky example as the points are on a manifold, and we are not considering their geometry. In this example, both algorithms perform similarly in identifying two of the three anomalies, but also producing several false positives. The overall performance of the new and old versions of lookout is similar.

4.6 Real-world Examples

We check the new and old versions of lookout on two sets of real data. Neither of these examples have labeled anomalies and so we cannot compute accuracy measures. However, as both datasets have a small number of variables, we can visualize the output. The datasets are

1. Old faithful dataset¹: eruptions of the Old Faithful Geyser in Yellowstone National Park, Wyoming, USA, from 1 January 2015 to 1 October 2021. For each recorded eruption the duration is given in seconds along with the waiting time to the following eruption. The records are incomplete, especially during winter.
2. Wine reviews dataset²: different attributes of wines from 44 countries, obtained from the *Wine Enthusiast Magazine* during the week of 15 June 2017. We focus on the numerical points score given by the *Wine Enthusiast* reviewer on a scale of 0–100 and the corresponding price of each bottle of wine.

Fig. 6 shows the output of new and old versions of lookout when applied to the *Old Faithful* dataset, with the identified anomalies depicted in red. New lookout identifies fewer “anomalous” points, especially in the region between the two clouds of points, while identifying some additional “anomalies” on the edge of the largest cloud of points. It is not clear which version is to be preferred in this case.

Fig. 7 shows the output of lookout on the *Wine Reviews* dataset. For this dataset we see that the new version identifies far fewer anomalies compared the old version, especially at the extremes of the points

¹Data downloaded from <https://geysertimes.org>.

²Data downloaded from <https://www.kaggle.com/datasets/zynicide/wine-reviews/data>.

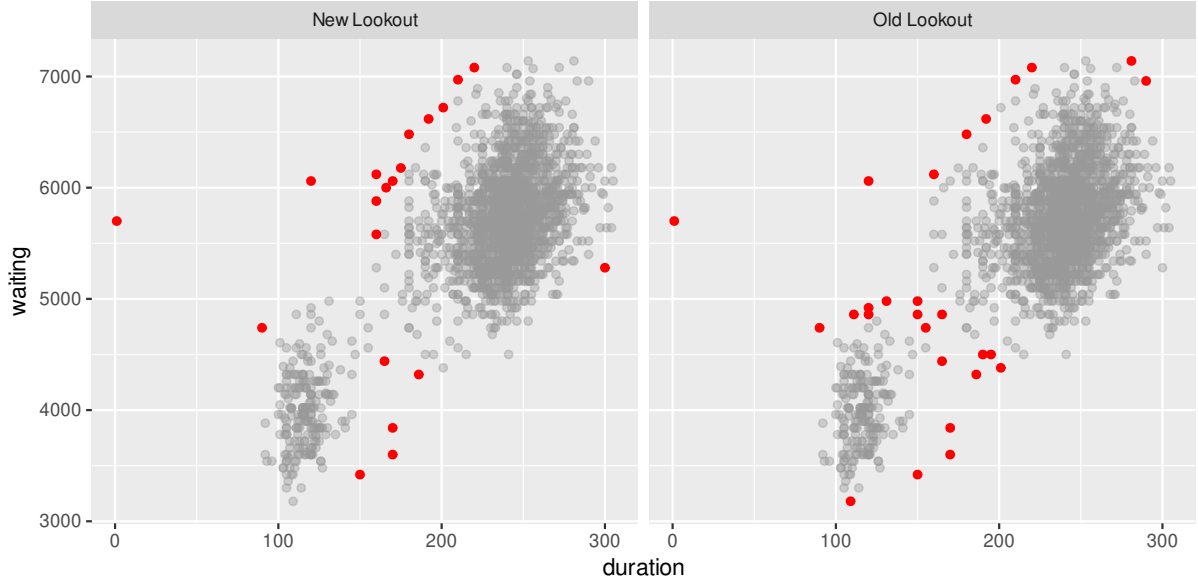


Fig. 6: Anomalies in the Old Faithful dataset

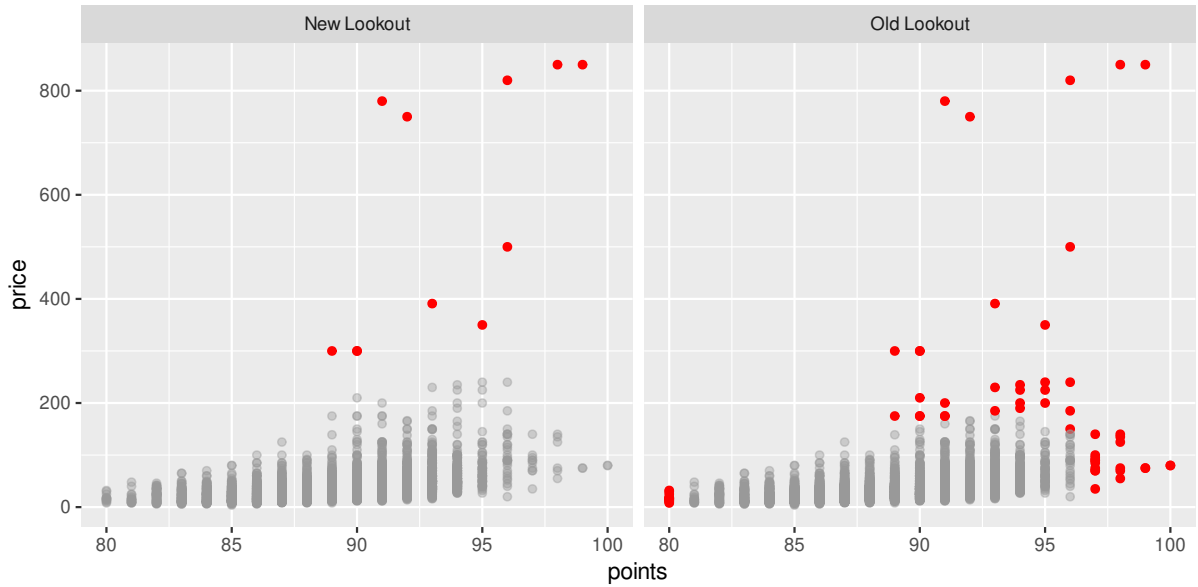


Fig. 7: Anomalies in the Wine Reviews dataset

score, and around the main cloud of points. In this example the new lookout algorithm is a substantial improvement over the old.

5 Evaluating lookout with other anomaly detection methods

In this section we compare the new and old versions of lookout with two Extreme Value Theory (EVT) based methods and two KDE-based methods. The EVT-based methods are HDoutliers ([Wilkinson 2017](#)) and *stray* ([Talagala et al. 2020](#)), both with their α parameter set to zero (which is the default for *stray*). The KDE-based methods are KDEOS ([Schubert et al. 2014](#)) and RDOS ([Tang and He 2017](#)), both run

with their default settings. The same anomaly detection methods were used for comparison in the original version of lookout ([Kandanaarachchi and Hyndman 2022](#)).

Both HDoutliers and stray identify anomalies, i.e., they give a binary label 1 to anomalies and 0 to non-anomalies. Therefore, to compare lookout with stray and HDoutliers we have used the metrics *Fmeasure* and *Gmean*. Suppose TP denotes the true positives (actual = 1, predicted = 1), FP denotes the false positives (actual = 0, predicted = 1), TN denotes the true negatives (actual = 0, predicted = 0) and FN denotes the false negatives (actual = 1, predicted = 0). Then these metrics are defined as follows:

$$Fmeasure = \frac{2(Precision \times Recall)}{Precision + Recall} \quad \text{where} \quad Precision = \frac{TP}{TP + FP} \quad \text{and} \quad Recall = \frac{TP}{TP + FN},$$

$$Gmean = \sqrt{Sensitivity \times Specificity} \quad \text{where} \quad Sensitivity = \frac{TP}{TP + FN} \quad \text{and} \quad Specificity = \frac{TN}{TN + FP}.$$

Note that recall and sensitivity define the same quantity. Sensitivity is commonly used in epidemiological settings while recall is used in computer science.

The KDE-based methods KDEOS and RDOS give anomaly scores instead of identifying points as anomalous or not. A standard way to compare anomaly scores is by using the area under the Receiver Operator Curve (ROC). For a given set of anomaly scores the ROC plots 1 - Specificity on the x -axis for different cut-off points and Sensitivity on the y -axis. A method A is better than method B if the area under the ROC is greater for method A compared to B . As lookout gives anomaly scores as well as binary labels, we compare lookout with KDEOS and RDOS using the area under the ROC, which we denote by AUC.

5.1 Experiment 5: Normal distributions moving out in each iteration

In this experiment we consider 405 points in $\mathbb{R}^6$, of which 5 points are anomalies. The experiment is run for 10 iterations with anomalies becoming more obvious in each iteration. The non-anomalies are distributed as $\mathcal{N}(0, 1)$ in all dimensions. The anomalies differ the non-anomalies only in the first dimension. The anomalies are distributed as $\mathcal{N}(2 + (i - 1) \times 0.5, 0.2)$ for iterations $i \in \{1, \dots, 10\}$ in the first dimension. In the other dimensions the anomalies are distributed as $\mathcal{N}(0, 1)$. We run the experiment for 10 iterations and in each iteration the anomalies move further away from the non-anomalous points. Each iteration is repeated 10 times to account for randomness.

[Fig. 8](#) shows the data and the results of this experiment, with anomalies shown in red. The top left plot shows the data on the Y_1Y_2 plane for the 3rd iteration and the middle two plots show the data for the 9th iteration. We see that the anomalies have separated from the other points in the Y_1 direction, but mixed with the non-anomalies on the Y_3Y_4 plane.

The plot on the right gives the AUC when comparing lookout with KDEOS and RDOS and *Fmeasure* and *Gmean* for comparison with HDoutliers and stray. As discussed the reason for different metrics is

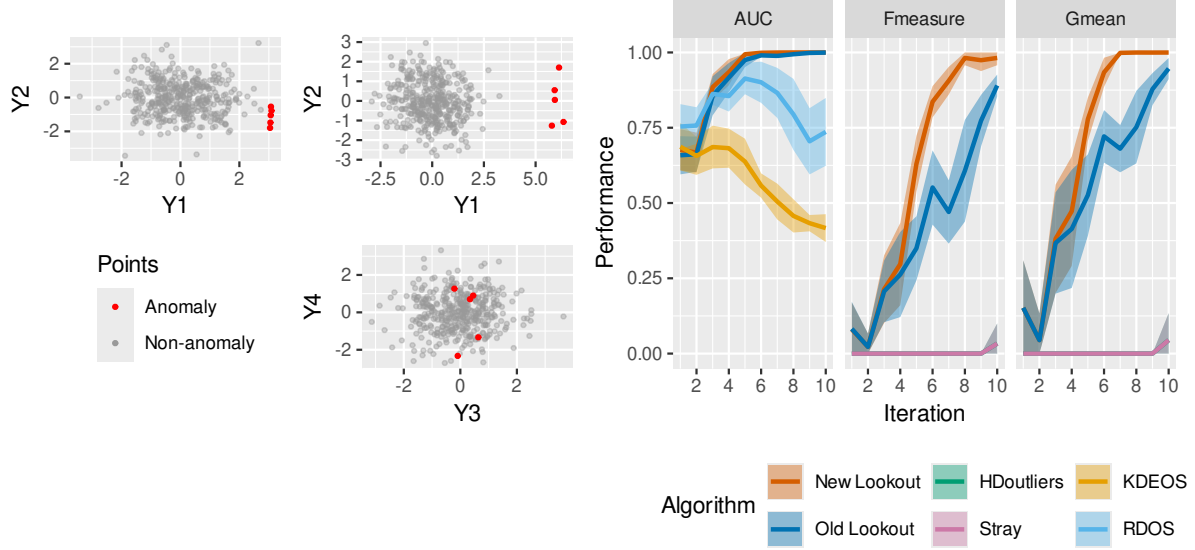


Fig. 8: Experiment 5 results. Top left: Data from the 3rd iteration showing the Y_1Y_2 plane with anomalies in red. Middle left: Data from the 9th iteration showing both the Y_1Y_2 plane and Y_3Y_4 plane, with anomalies moving further away in the Y_1 direction, resulting in the anomalies and non-anomalies being mixed on the Y_3Y_4 plane. Right: AUC, Fmeasure and Gmean of old and new versions of lookout, HDoutliers, KDEOS, RDOS and stray.

because KDEOS and RDOS give anomaly scores, while stray and HDoutliers identify anomalies. We see that new lookout performs better than the old lookout in terms of Gmean and Fmeasure for higher iterations. In terms of AUC both versions of lookout give the same performance.

5.2 Experiment 6: Annulus in $\mathbb{R}^4$

For this experiment we consider 805 points in $\mathbb{R}^3$ of which 5 points are anomalies. Fig. 9 shows the data lying in an annulus on the Y_1Y_2 plane with $Y_3 \sim \mathcal{U}(0, 1)$. As before, the experiment is run for 10 iterations with anomalies moving towards the centre of the hole in each iteration. Plots from iteration 3 and 9 are shown on the left in Fig. 9. The results of the different methods are shown on the right.

5.3 Experiment 7: Unit cube in $\mathbb{R}^{20}$

This experiment considers 500 points in $\mathbb{R}^{20}$ of which one is an anomaly. At the start of the experiment all 500 points are uniformly distributed with $y_i \sim \mathcal{U}(0, 1)$ in each dimension. For this experiment we consider 20 iterations and in each iteration the 500th data point becomes more anomalous. In the first iteration the 500th point is updated with the first coordinate $Y_1 = 0.9$. In the second iteration it is further updated with $Y_2 = 0.9$, resulting in $(0.9, 0.9, Y_3, \dots, Y_{20})$. In the i th iteration the 500th point is updated with $Y_i = 0.9$. Each iteration is repeated 10 times to account for randomness.

The anomalous point is not anomalous in any single dimension or on any Y_iY_j plane. In fact even in the space of the first two principal components, the anomaly is not distinguishable. We use *dobin*

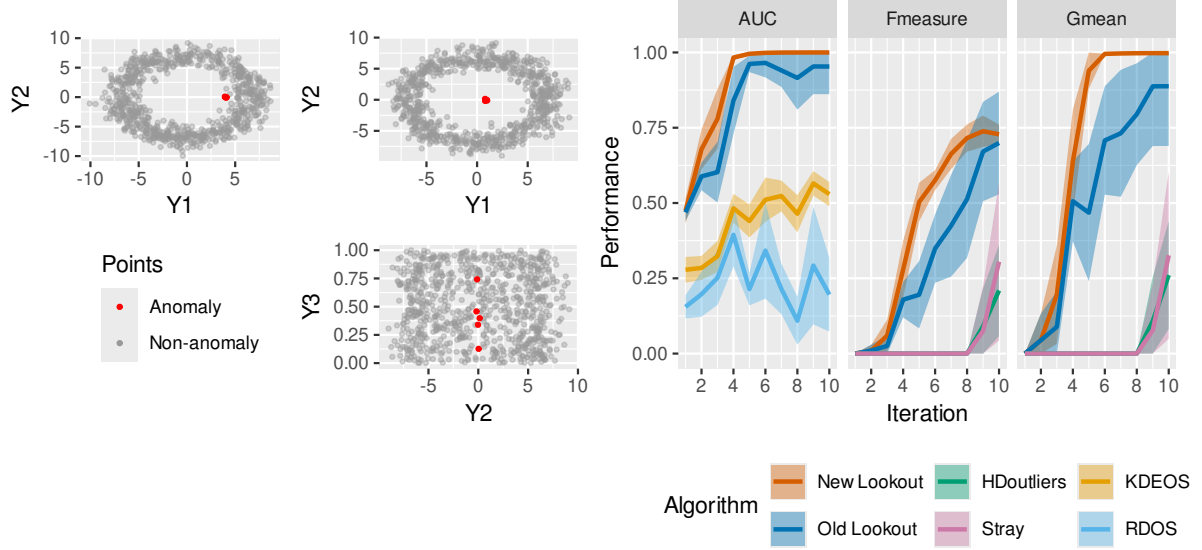


Fig. 9: Experiment 6 results. Top left: Data from the 3rd iteration showing the Y_1Y_2 plane with anomalies in red. Middle left: Data from the 9th iteration showing both the Y_1Y_2 plane and Y_2Y_3 plane, with anomalies moving into the annulus. Right: AUC, Fmeasure and Gmean of old and new versions of lookout, HDoutliers, KDEOS, RDOS and stray.

(Kandanaarachchi and Hyndman 2021), a dimension reduction method especially developed for anomalies to visualize this space. Fig. 10 shows the data and the results of this experiment. The plots on the left show the data on the D_1D_2 plane for iterations 5, 12 and 20, where D_1 and D_2 are the first two axes of dobin projection. We see that in the 20th iteration the anomaly can be easily identified on the D_1D_2 plane. The plot on the right shows the results of applying the various algorithms to this problem. The lookout algorithms are applied without scaling in this experiment.

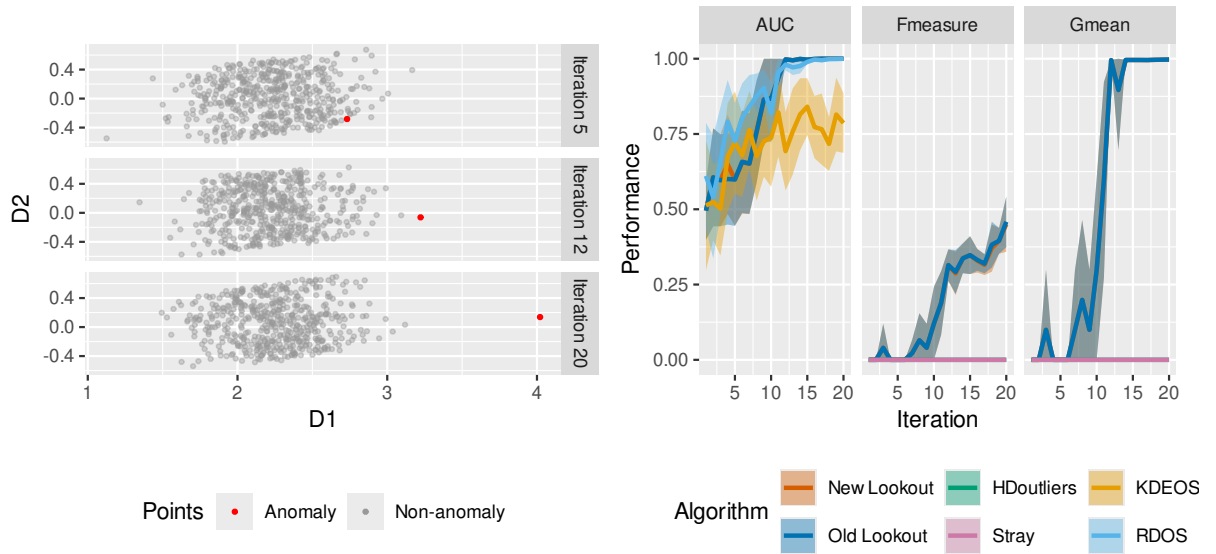


Fig. 10: Experiment 7 results. Left: Data from the 5th, 12th and 20th iterations, showing the D_1D_2 plane from the dobin projection. Anomalies are shown in red. Right: AUC, Fmeasure and Gmean of old and new versions of lookout, HDoutliers, KDEOS, RDOS and stray.

6 Discussion and conclusions

We have proposed an improved version of the lookout algorithm based on theoretical underpinnings in persistent homology. The updated algorithm uses an upper percentile of the Rips death diameters rather than the maximum successive difference for bandwidth selection, which we prove yields a consistent kernel density estimator under broad conditions. Additionally, we replace min-max scaling with robust multivariate scaling using the orthogonalized Gnanadesikan-Kettenring estimator, improving robustness to outliers and accounting for correlations between variables. Finally, we constrain the shape parameter of the Generalized Pareto Distribution to be non-positive when fitting to the negative log kernel density estimates, leveraging known bounds on these values to enhance stability.

Our empirical evaluation demonstrates that the updated lookout performs better than the original version across diverse examples, while maintaining competitive performance against other anomaly detection methods. The algorithm provides probabilistic anomaly scores with theoretical grounding in extreme value theory, enabling principled threshold selection.

The key strengths of the updated lookout algorithm include: (1) theoretical consistency guarantees for the kernel density estimator under mild conditions; (2) robustness to outliers in both the scaling and bandwidth selection steps; (3) principled probabilistic interpretation through extreme value theory; and (4) automatic adaptation to the local topological structure of the data through persistent homology.

The algorithm is versatile and suited to any continuous multivariate data, and can handle moderate dimensionality effectively (the experiments show good results up to about 20 dimensions) effectively. It can also be applied to any problems that can be transformed into a continuous multivariate setting, such as image anomaly detection via feature extraction, or functional data anomaly detection via functional principal components. We also assume that the data are independent and identically distributed, making it unsuitable for temporal or spatial data without pre-processing (e.g., applying it to the residuals of a fitted model).

However, several computational limitations remain. The computational complexity is dominated by the computation of dimension zero Vietoris-Rips persistent homology, which is equivalent to single link clustering, requiring $O(n \log n)$ time (Gower and Ross 1969). This can be prohibitive for very large datasets, although approximate methods or subsampling could mitigate this. The method may also struggle with very high-dimensional data due to the curse of dimensionality affecting kernel density estimation. Using dimension reduction methods, such as principal component analysis or dobin (Kandanaarachchi 2019), before applying lookout could help in such cases.

Supplementary information. The updated algorithm is implemented in the lookout R package (Kandanaarachchi et al. 2025), available on CRAN at <https://CRAN.R-project.org/package=lookout>.

Files to reproduce all numerical examples are available at <https://github.com/robjhyndman/Look-Out-2>

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